

	1	2	3	4	5	
A	12V Power Input File: POS12_power_input.kicad_sch 12V Telemetry File: POS12_telemetry.kicad_sch 3.3V Power Supply File: POS3P3_power_supply.kicad_sch 3.3V Telemetry File: POS3P3_telemetry.kicad_sch dsPIC33 Programming File: dspic33_programming.kicad_sch dsPIC33 Power File: dspic33_power.kicad_sch dsPIC33 GPIO File: dspic33_gpio.kicad_sch USB UART Bridge File: usb_uart_bridge.kicad_sch		Platform ETC File: platform_etc.kicad_sch PGOOD LEDs File: pgood_leds.kicad_sch Status LEDs File: status_leds.kicad_sch RCA Signal Conditioning File: rca_signal_conditioning.kicad_sch Speaker Signal Conditioning File: speaker_signal_conditoning.kicad_sch Source Selection, PGAs File: source_selection_pgas.kicad_sch Anti-alias Filtering File: anti_alias_filtering.kicad_sch Audio Voltage References File: audio_voltage_references.kicad_sch LED Panel Connectors File: led_panel_connectors.kicad_sch Pushbuttons File: pushbuttons.kicad_sch Mechanical File: mechanical.kicad_sch			
B						
C						
D						
	1	2	3	4	5	

Sheet: / File: Audio_Visualiser_DSP.kicad_sch		
Title:		
Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 1/20

[illegible]

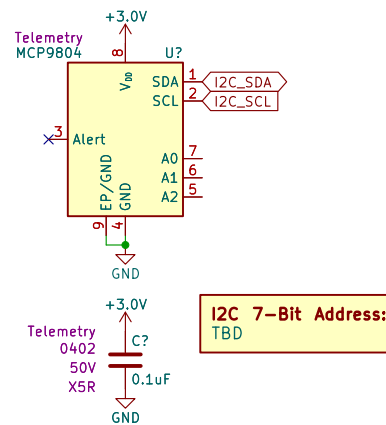
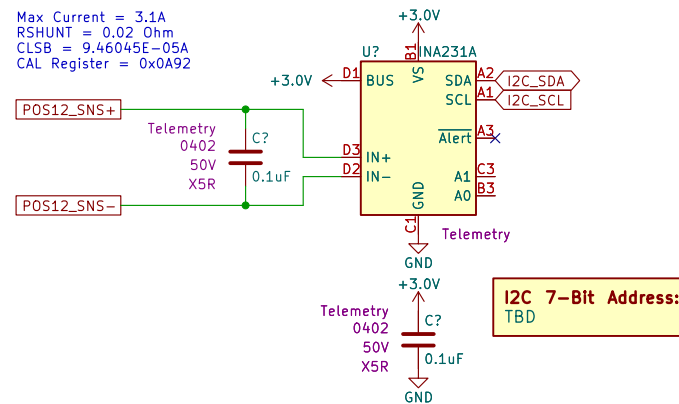
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Size: A	Date: 2023-12-25	Rev:
KICad E.D.A. 10.0.3		Id: 2/20

Size: A	Date: 2023-12-25	Rev:
KiCad E.D.A. 10.0.3		Id: 2/20

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[illegible]

Rev:
Id: 2/20



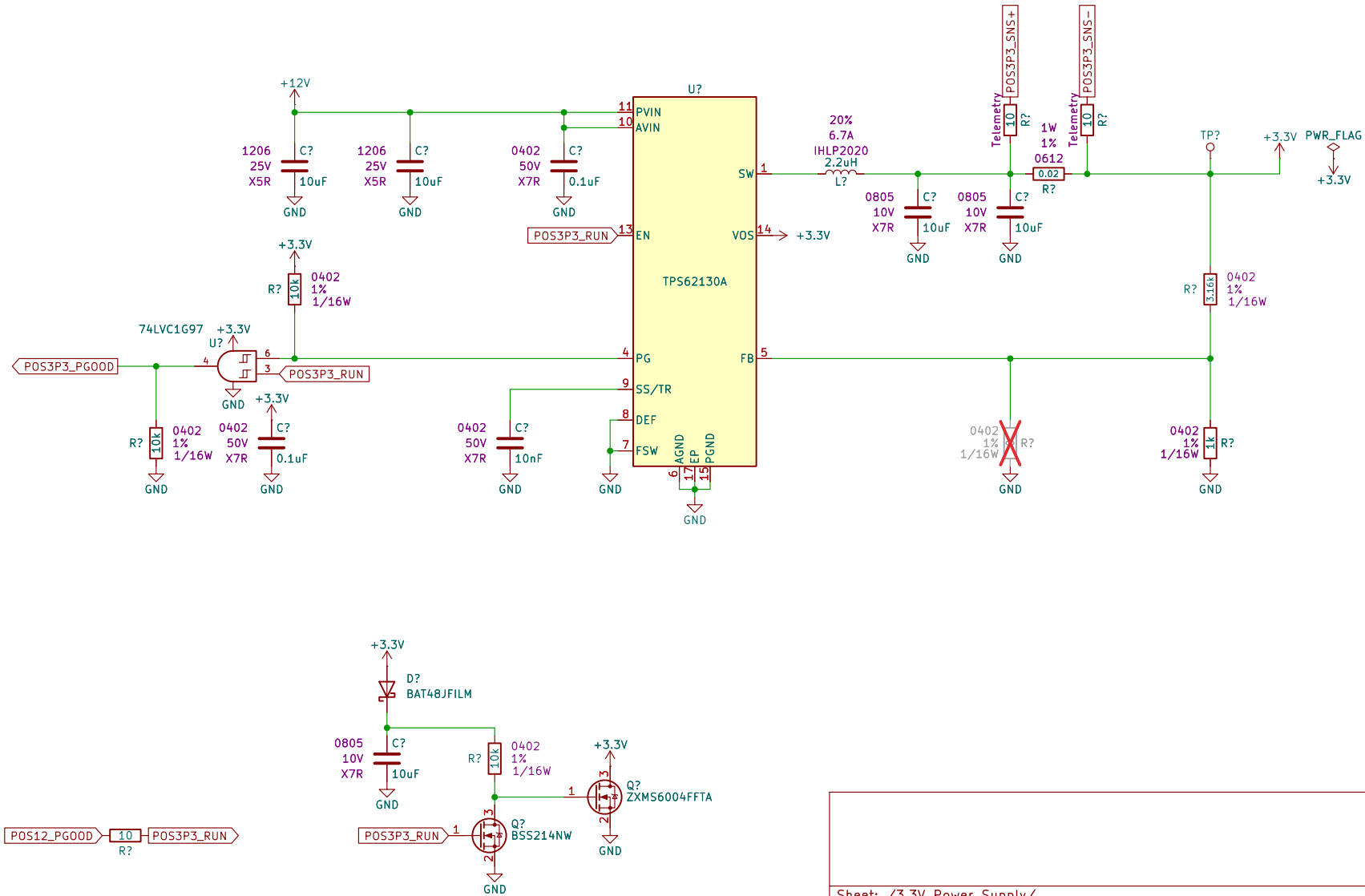
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 File: POS12_telemetry.kicad_sch

Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 3/20

TRY TO TALLY LOAD CURRENTS



Sheet: /3.3V Power Supply/
File: POS3P3_power_supply.kicad_sch

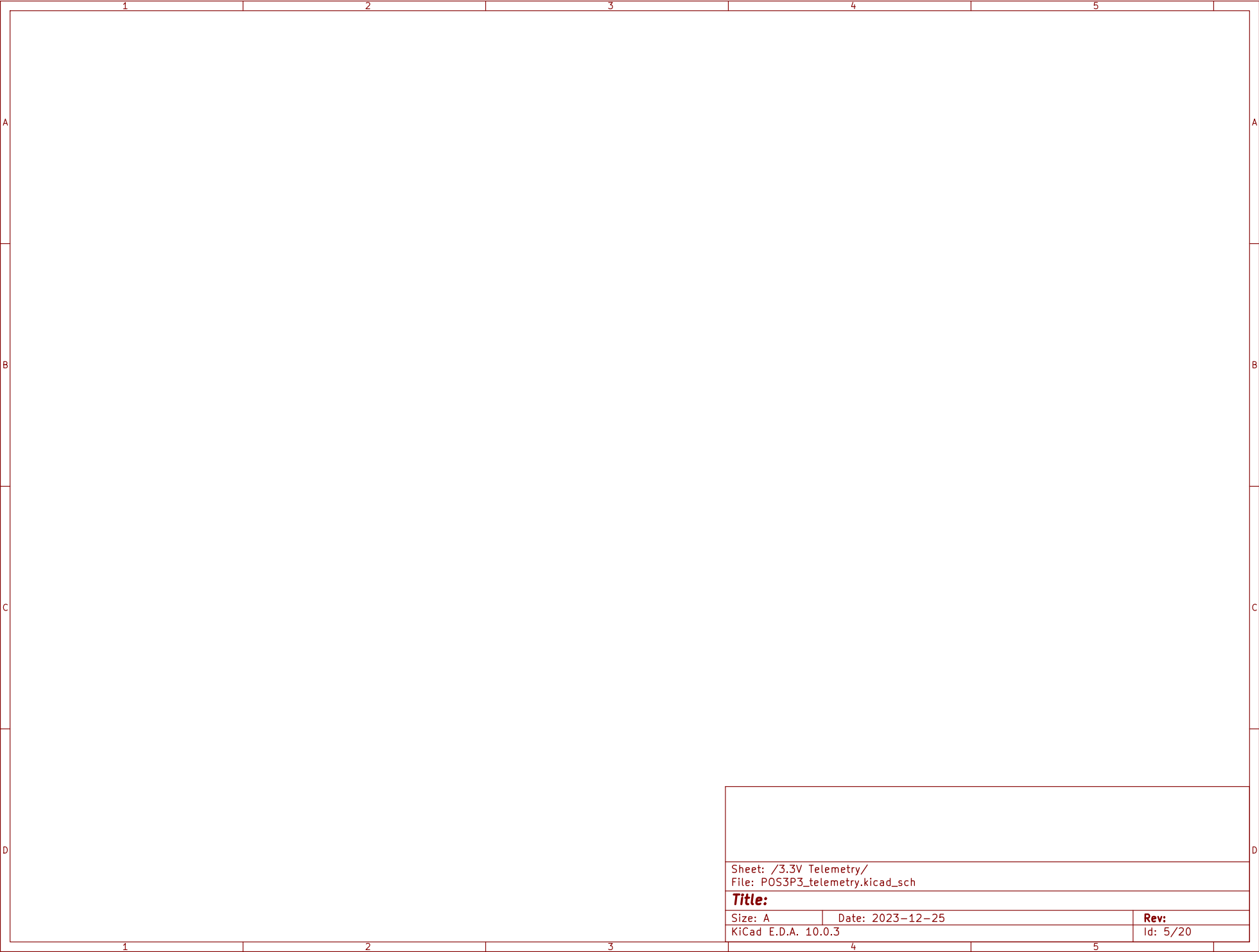
Title:

Size: A	Date: 2023-12-25
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KiCad E.D.A. 10.0.3

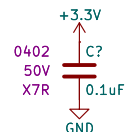
Rev:

Id: 4/20



I2C 7-Bit Address:

ADD HARDSTRAPS FOR CONFIG STUFF



U?B

dsPIC33AK256MPS105

OA1IN+/AD1AN1

ADC_AUDIO_LEFT

AUDIO_VREF_ENABLE

OA2IN+/AD2AN1

SDA1/PGD1

SCL1/PGC1

RP23 = REF01

ADC_AUDIO_RIGHT

DSPIC33_ICSPDAT

DSPIC33_ICSPCLK

DSPIC33_REFCLK1

OSCO/CLK0/RP33/IOMAF5/RC0

OSCI/RP34/IOMAF6/RC1

PGC3/RP35/PWM4H/IOMAD7/RC2

PGD3/RP38/I3CSDA1/PWM3L/IOMAD6/RC5

RP37/I3CSDA1/PWM3L/IOMAD4/RC4

RP39/ASCL1/RC6

RP40/ASDA1/IOMAF7/RC7

RP49/PWM2H/IOMAD3/RD0

TCK/RP50/PWM2L/IOMAD2/RD1

TDI/RP52/PWM1L/IOMAD0/RD3

RP55/RD6

USB_UART_RX

DSPIC33_POSC_EC

USB_UART_TX

LED_DRIVER_BLANK_PWM

Heartbeat_LED

LED_DRIVER_LAT

LEFT_LED_DRIVER_SPL_SCK

LEFT_LED_DRIVER_SPL_SDO

RIGHT_LED_DRIVER_SPL_SCK

RIGHT_LED_DRIVER_SPL_SDO

RP33 = U1RX

OSCI

RP35 = U1TX

RP36 = OCM2

RP37 = OCM1

RP49 = SCK1

RP50 = SDO1

RP51 = SCK2

RP52 = SDO2

Enable internal pullups in firmware

Internal 250k pullup to VCCB

Isolate I2C signals when in reset/being programmed so that ICSP can program the DSC but cannot spook I2C devices with nonsensical data

Pulled high when in reset via resistor on status LEDs page

Low when in reset

Sheet: /dsPIC33 GPIO/
File: dspic33_gpio.kicad_sch

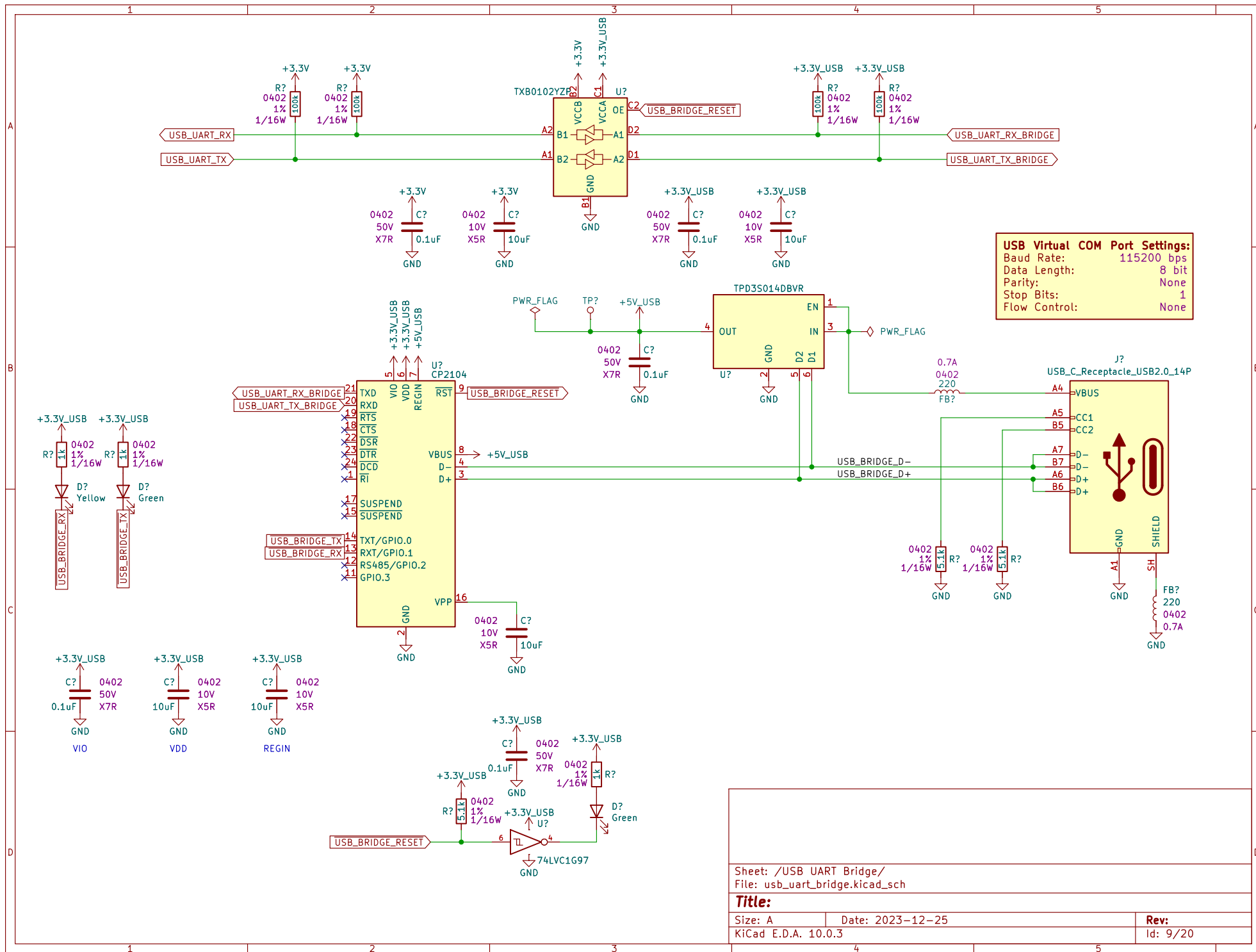
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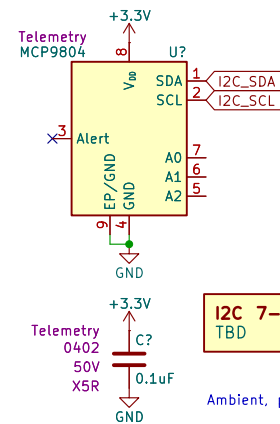
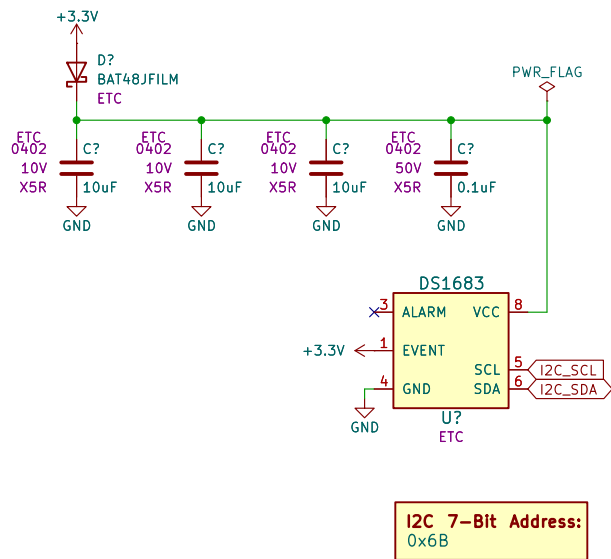
Size: A Date: 2023-12-25

KiCad E.D.A. 10.0.3

Rev:

Id: 8/20





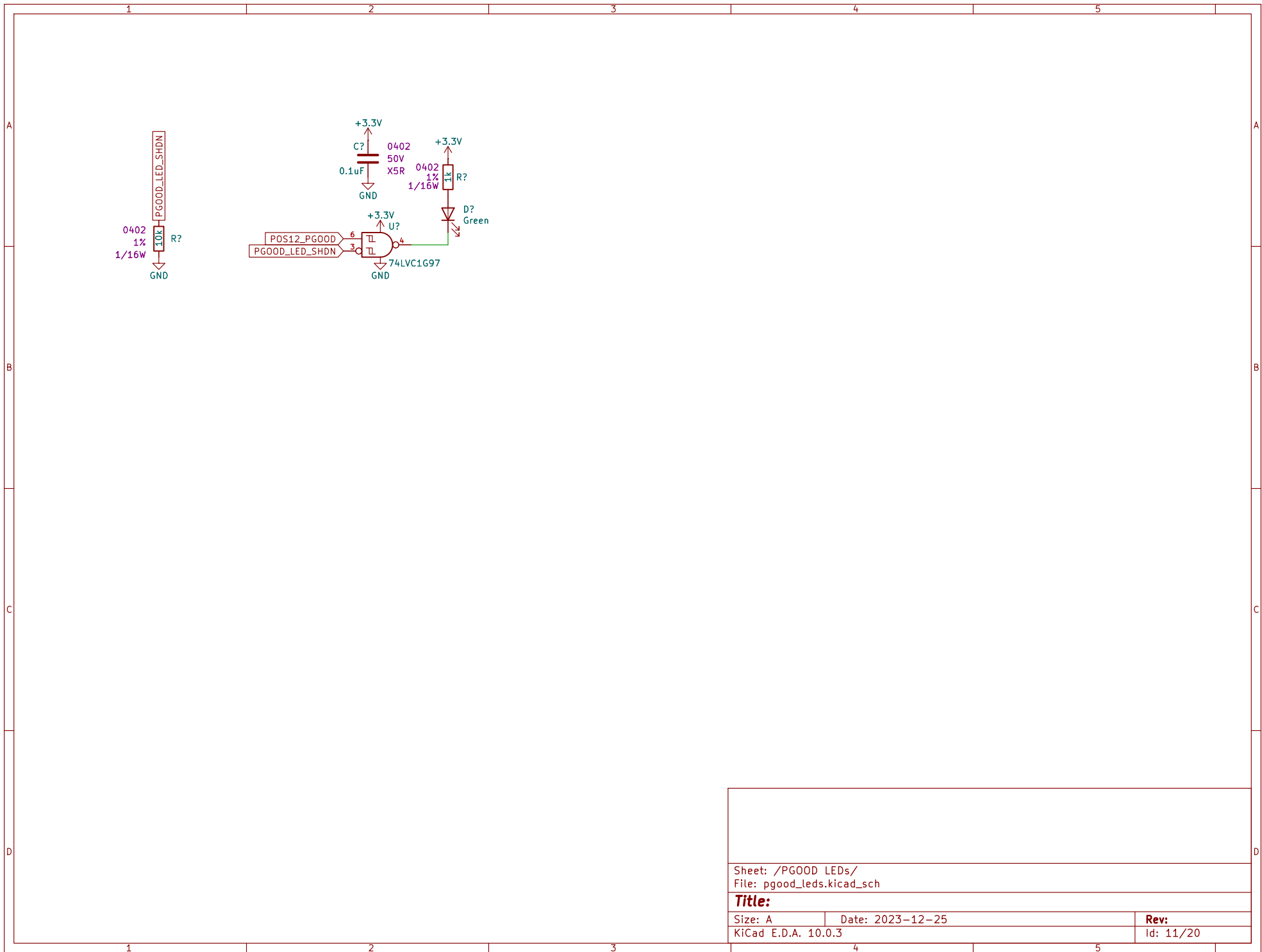
Ambient, place away from power components

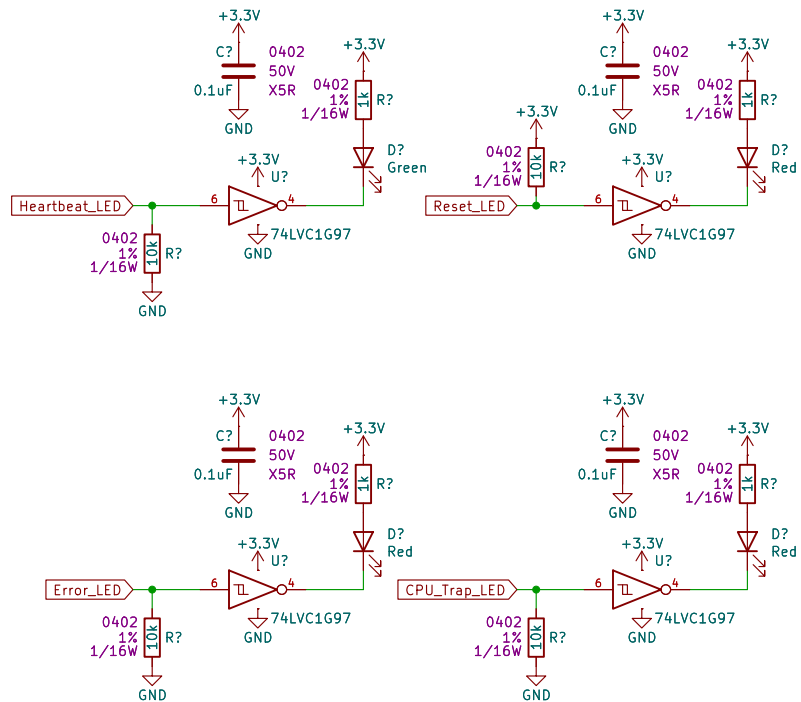
Sheet: /Platform ETC/
File: platform_etc.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 10/20



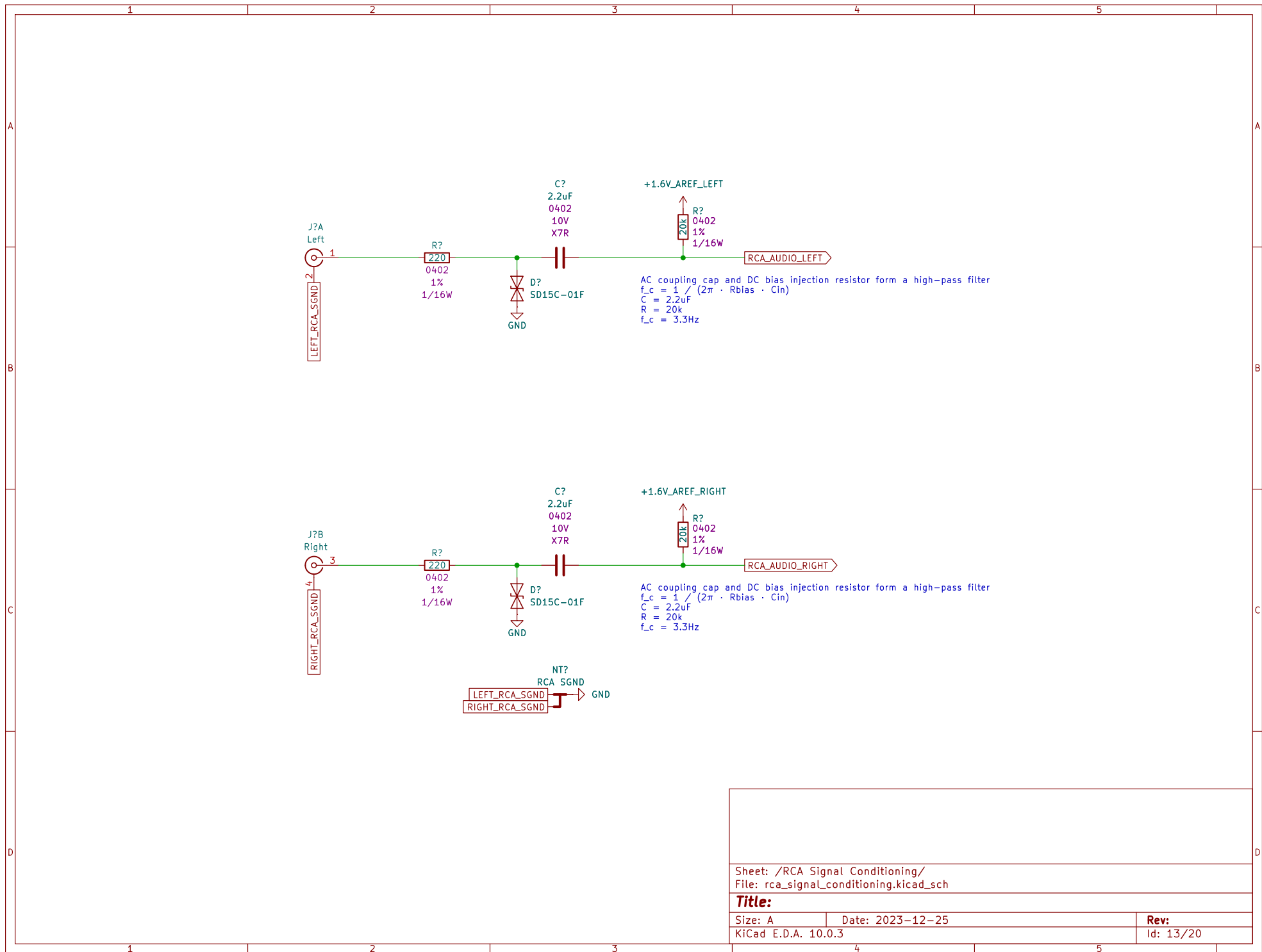


Sheet: /Status LEDs/
File: status_leds.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 12/20



Sheet: /RCA Signal Conditioning/
File: rca_signal_conditioning.kicad_sch

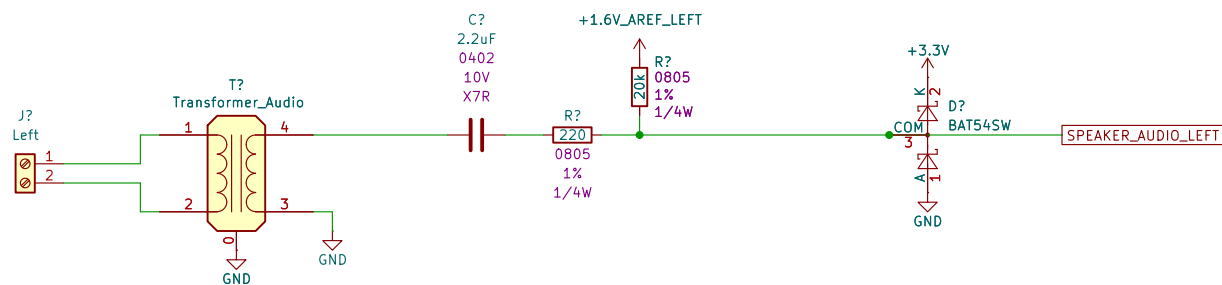
Title:

Size: A Date: 2023-12-25

KiCad E.D.A. 10.0.3

Rev:
Id: 13/20

Alternate is a 20:1 resistive attenuation (for 18V peaks) and a diff amp



Sheet: /Speaker Signal Conditioning/
File: speaker_signal_conditioning.kicad_sch

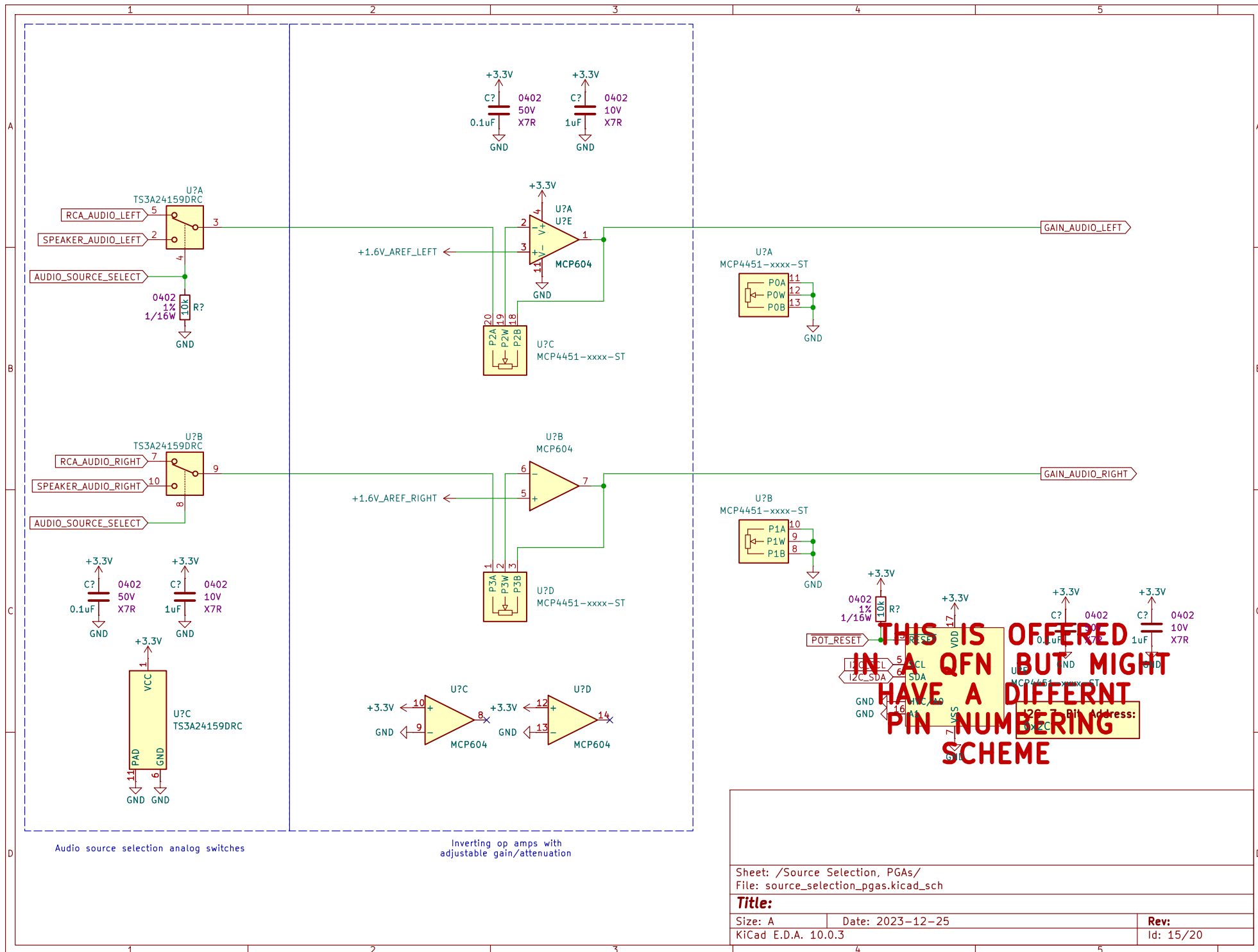
Title:

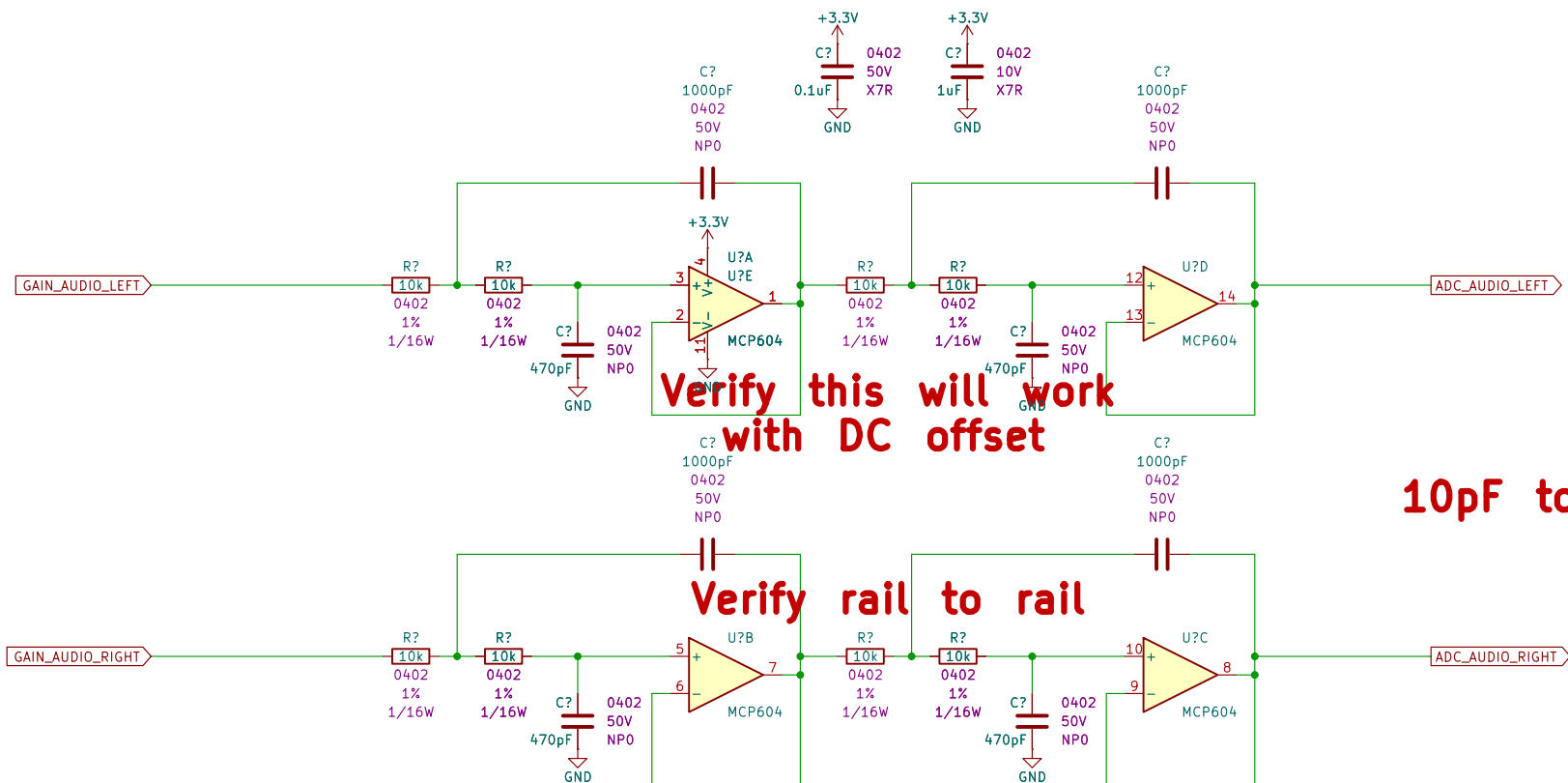
Size: A Date: 2023-12-25

KiCad E.D.A. 10.0.3

Rev:

Id: 14/20





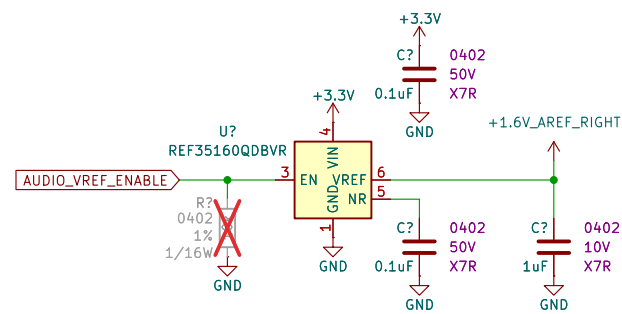
4-pole Sallen-Key active anti-aliasing lowpass filters
 Unity gain
 $f_c = 23.2\text{kHz}$, $Q = 0.707$ each
 If sampling at 44.1kHz , nyquist is 22.05kHz
 Must filter hard just below this

Sheet: /Anti-alias Filtering/
 File: anti_alias_filtering.kicad_sch

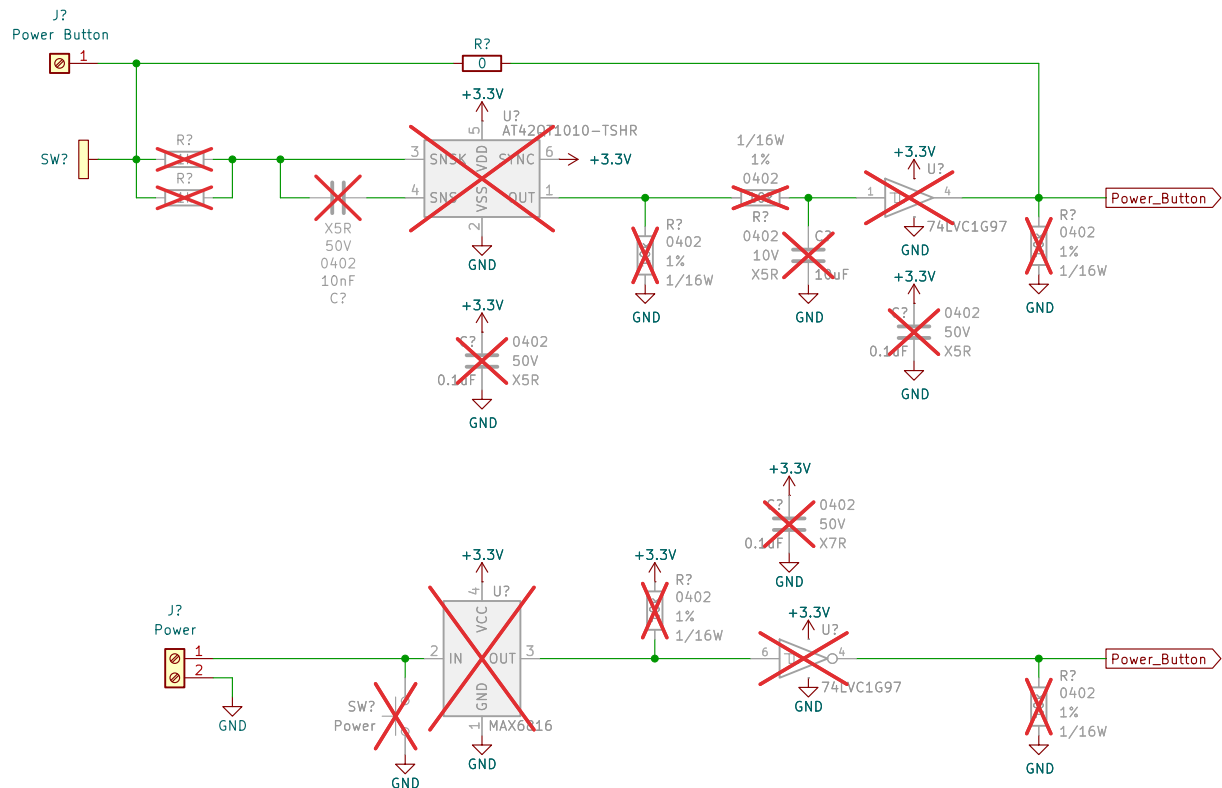
Title:

Size: A Date: 2023-12-25
 KiCad E.D.A. 10.0.3

Rev:
 Id: 16/20



Rev:
Id: 17/20



Sheet: /LED Panel Connectors/
File: led_panel_connectors.kicad_sch

Title:

Size: A
KiCad E.D.A. 10.0.3

Date:

Rev:
Id: 18/20

1					2					3					4					5				
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B																								
C																								
D																								
1					2					3					4					5				

Sheet: /Pushbuttons/
File: pushbuttons.kicad_sch

Title:

Size: A

Date:

Rev:

KiCad E.D.A. 10.0.3

Id: 19/20

Sheet: /Pushbuttons/ File: pushbuttons.kicad_sch																								
Title:																								
Size: A					Date:															Rev:				
KiCad E.D.A. 10.0.3																				Id: 19/20				

1					2					3					4					5					
A																									A
B																									B
C																									C
D																									D
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Sheet: /Mechanical/
File: mechanical.kicad_sch

Title:

Size: ADate:KiCad E.D.A. 10.0.3

Rev:Id: 20/20

Sheet: /Mechanical/ File: mechanical.kicad_sch																								
Title:																								
Size: A					Date:															Rev:				
KiCad E.D.A. 10.0.3					Id: 20/20																			